Deep Phylogeny

Theoretical foundations for sequence-based and protein structural phylogenetics
Workshop theory handout

Purpose of this handout

This document explains the concepts behind the practical workflow used during the workshop. It is designed to help participants understand what each algorithm is trying to infer, which assumptions are being made, and where uncertainty enters the analysis. It is not a substitute for a full course in molecular evolution.

Core workflow

Stage	Question	Main tools
Dataset construction	Which homologues should be compared?	UniProt; taxon-guided retrieval
Domain definition	Are we comparing homologous regions?	InterPro; InterProScan; ProtDomRetriever
Structure inspection	Are reliable structural models available?	AlphaFold DB
Alignment	Which residues are positionally homologous?	MAFFT; MUSCLE 5; Jalview
Alignment assessment	How robust is the positional-homology hypothesis?	MAFFT_ScoreNGo
Sequence phylogeny	Which tree best explains the alignment under an evolutionary model?	IQ-TREE
Structural phylogeny	What evolutionary signal is retained in protein structure?	Foldseek; FoldTree
Interpretation	Which evolutionary scenarios are consistent with all evidence?	iTOL; FigTree; biological context

1. Phylogeny is a model-based inference

A phylogenetic tree is a hypothesis about historical relationships. For protein families, we do not observe the ancestral proteins directly. Instead, we observe present-day sequences and structures and ask which evolutionary history provides a plausible explanation for those observations.

This distinction matters because the final tree depends on several layers of inference: which proteins were selected, which residues were aligned, which evolutionary model was used, how the tree space was searched, and how uncertainty was quantified.

Key idea

A sophisticated tree-building algorithm cannot rescue a biologically inappropriate dataset or a badly misaligned homologous region.

2. Start with the biological question, not the software

Before collecting sequences, define the evolutionary question. Examples include: When did a protein family diversify? Did a domain appear before or after a particular lineage split? Is an apparent similarity more consistent with vertical inheritance, duplication, horizontal transfer, domain shuffling or convergence?

The question determines the sampling. A taxonomically balanced dataset is often more informative than a large dataset dominated by a few well-sequenced clades.

- Include taxa that distinguish between the competing evolutionary hypotheses.
- Avoid unnecessary redundancy from hundreds of nearly identical sequences.
- Consider an outgroup when a biologically defensible outgroup exists.
- Be alert to lineage-specific gene duplications and paralogues.
- For multidomain proteins, decide whether the whole protein or a shared domain is the true unit of comparison.

3. Homology, orthology and paralogy

Homology means common ancestry. Two proteins are homologous or they are not; strictly speaking, **'percent homology' <u>is not</u> meaningful.** This error is still encountered surprisingly often, even in published papers and oral presentations, and is particularly common among students. It is incorrect: **two proteins are either homologous or they are not.** Homology itself is not a percentage, although our confidence in inferring homology can vary. In multidomain proteins, homology may also apply to a specific domain or region rather than to the entire protein.

<u>Sequence identity and sequence similarity are measurable quantities</u>, whereas homology is an evolutionary relationship inferred from evidence.

Orthologues are homologues separated by a speciation event. Paralogues are homologues separated by a duplication event. A protein family can contain both, and confusing them can lead to incorrect biological conclusions even when the inferred tree itself is technically correct.

Common pitfall

A taxonomic label does not guarantee orthology. One species can contain several paralogues, and the most similar sequence is not always the orthologue relevant to your question.

<u>Good reading:</u> Orthologs, Paralogs, and Evolutionary Genomics – E. V. Koonin
[10.1146/annurev.genet.39.073003.114725](https://doi.org/10.1146/annurev.genet.39.073003.114725)

XGD = xenologous gene displacement, i.e. replacement of a resident gene by a horizontally transferred homolog. Two species can therefore retain homologous genes that have different evolutionary histories.

Table 1 Homology: terms and definitions

Homologs	**Genes sharing a common origin**
Orthologs	Genes originating from a single ancestral gene in the last common ancestor of the compared genomes.
Pseudoorthologs	Genes that actually are paralogs but appear to be orthologous due to differential, lineage-specific gene loss.
Xenologs	Homologous genes acquired via XGD by one or both of the compared species but appearing to be orthologous in pairwise genome comparisons.
Co-orthologs	Two or more genes in one lineage that are, collectively, orthologous to one or more genes in another lineage due to a lineage-specific duplication(s). Members of a co-orthologous gene set are inparalogs relative to the respective speciation event.
Paralogs	**Genes related by duplication**
Inparalogs (symparalogs)	Paralogous genes resulting from a lineage-specific duplication(s) subsequent to a given speciation event (defined only relative to a speciation event, no absolute meaning).
Outparalogs (alloparalogs)	Paralogous genes resulting from a duplication(s) preceding a given speciation event (defined only relative to a speciation event, no absolute meaning).
Pseudoparalogs	Homologous genes that come out as paralogs in a single-genome analysis but actually ended up in the given genome as a result of a combination of vertical inheritance and HGT.

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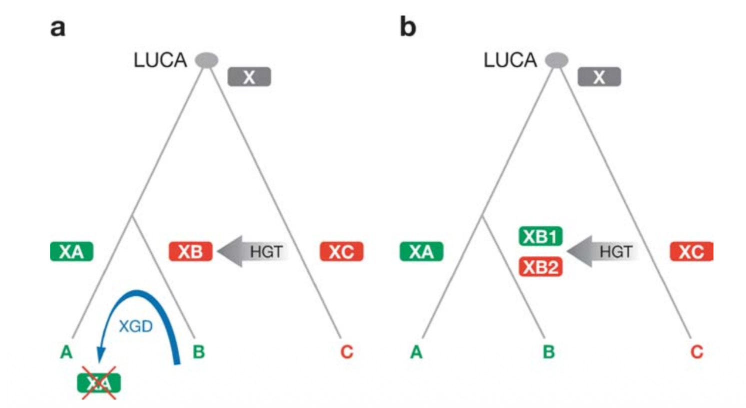


Figure 4
Effect of horizontal gene transfer on orthology and paralogy. (*a*) A hypothetical evolutionary scenario with HGT leading to xenology. (*b*) A hypothetical evolutionary scenario with HGT leading to pseudoparalogy. LUCA, Last Universal Common Ancestor (of all extant life forms).

4. UniProt and sequence provenance

UniProtKB provides protein sequences together with names, taxonomic information, functional annotation, evidence and cross-references. Reviewed Swiss-Prot entries are manually curated, whereas TrEMBL entries are computationally annotated and await full manual review.

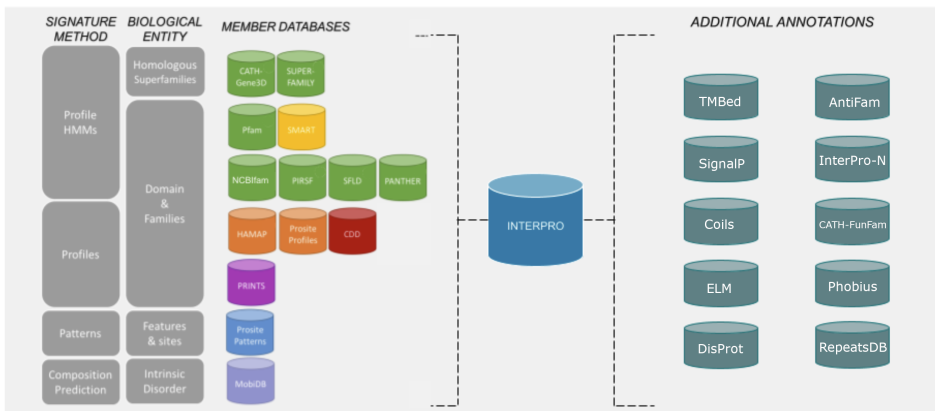
For phylogenetics, database provenance matters. Accession numbers should be retained throughout the workflow so that every sequence in the final tree can be traced back to its source.

- Check whether sequences are complete or fragmented.
- Check whether annotations are reviewed, predicted or inferred from similarity.
- Record the taxonomic lineage used during sampling.
- Keep a reproducible accession list rather than relying only on a FASTA file.

5. Protein domains and InterPro

Proteins are modular. A multidomain protein can gain, lose or rearrange domains while preserving one ancestral domain. If full-length proteins with different architectures are aligned, non-homologous regions can be forced into the same alignment and can distort the phylogenetic signal.

InterPro integrates predictive signatures from multiple member databases to classify proteins into families and identify domains, repeats, sites and other features. InterProScan applies the underlying signature methods to submitted sequences.



5.1 What is a profile HMM?

A profile hidden Markov model (HMM) is a statistical representation of a protein family or domain derived from a multiple sequence alignment. Instead of searching for one exact motif, a profile HMM models which residues are common at each position and where insertions or deletions are tolerated.

Conceptually, the model contains states corresponding to matches, insertions and deletions. A new sequence is scored according to how plausibly it could have been generated by that family-specific model.

Why this matters

Profile methods can detect family membership even when no single short sequence motif is perfectly conserved. Compared with pairwise similarity searches such as BLAST, profile HMMs use position-specific information derived from a multiple sequence alignment and are therefore often more sensitive for detecting remote homologues. BLAST, in contrast, performs local sequence-similarity searches between a query and database sequences.

5.2 Why extract a domain?

Domain extraction is appropriate when the evolutionary history of a conserved domain is the target of the analysis, especially when full-length proteins have variable architectures. However, domain extraction is not automatically superior: the biological question may genuinely concern the entire multidomain protein.

- Check that the same homologous domain is extracted in every sequence.
- Avoid mixing partial domains with complete domains.
- Inspect boundary uncertainty, especially in divergent proteins.
- Document the extraction rule so the analysis is reproducible.

5.3 Speaking the protein language

https://www.ebi.ac.uk/training/online/courses/protein-classification-intro-ebi-resources/protein-classification/

Learn how to answer the following questions:

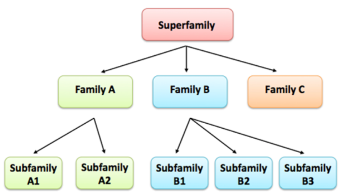


Figure 2 A hypothetical protein family hierarchy showing the relationships between superfamily, family and subfamily members. Directional arrows indicate that one group is a subgroup of another.

⇨ Why classify proteins?

- What are **protein families**?
- What are **protein domains**?
- What are **sequence features**?

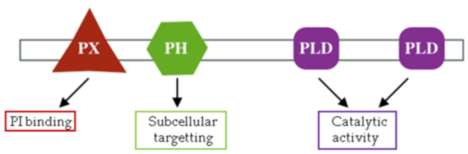


Figure 6 Domain composition of phospholipase D1, which is an enzyme that breaks down phosphatidylcholine.

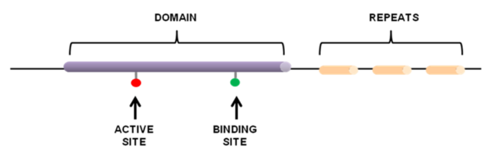


Figure 8 Graphical representation of repeats, domains and sites on a protein sequence.

Pattern/Motif

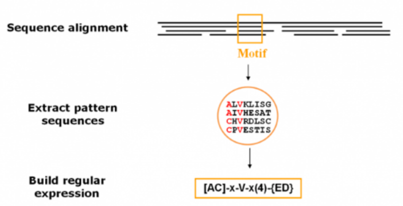


Figure 13 When creating patterns, a conserved motif is used to build a regular expression. The pattern illustrated here is translated as: [Ala or Cys]-any-Val-any-any-any-any-{any but Glu or Asp}.

Fingerprints

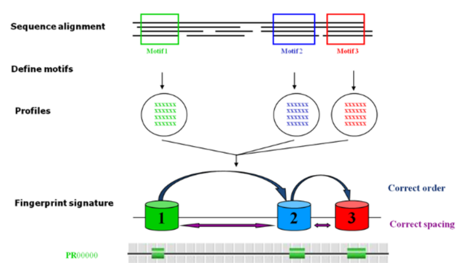


Figure 15 Representation of the steps involved in creating a fingerprint signature.

⇨ What are protein signatures?

- What are **patterns** (motifs)?
- What are **profiles**?
- What are **fingerprints**?
- What are **HMMs**?

Profiles

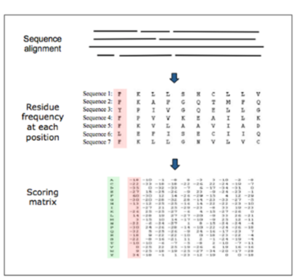


Figure 14 Representation of a scoring matrix based on a multiple sequence alignment. Each of the 20 amino acids commonly found in proteins is given a score for each position in the sequence according to the frequency with which they occur in the original alignment. Other factors, such as evolutionary distances can also be considered.

HMM

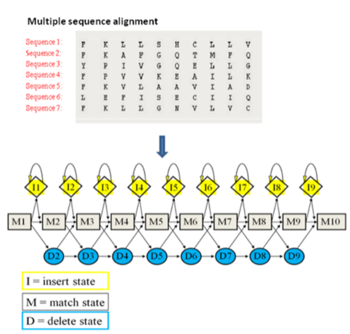


Figure 14 Representation of a Hidden Markov model based on a multiple sequence alignment. Amino acids are given a score at each position in the sequence alignment according to the frequency with which they occur. Transition probabilities (i.e., the likelihood that one particular amino acid follows another particular amino acid) and insertion and deletion states are also modelled.

6. AlphaFold structures: prediction and confidence

AlphaFold dramatically expanded structural coverage, making it possible to compare predicted structures for protein families that lack experimental structures. For evolutionary analysis, however, the coordinates must be interpreted together with confidence metrics.

The AlphaFold Protein Structure Database provides per-residue pLDDT confidence scores. High pLDDT generally indicates greater confidence in local geometry, whereas low values can reflect uncertainty, flexibility or intrinsic disorder. Predicted Aligned Error (PAE) is especially important for judging confidence in the relative placement of different regions or domains. Importantly, AlphaFold confidence scores estimate confidence in the predicted structure; they do not measure whether the predicted conformation is biologically relevant in a particular cellular or functional context.

Structural phylogenetics caution

A high-confidence fold can still be inappropriate for a particular evolutionary comparison if different proteins are being compared over non-equivalent domains or if flexible/low-confidence regions dominate the structural alignment.

7. Multiple sequence alignment: a hypothesis of positional homology

A multiple sequence alignment (MSA) proposes that residues placed in the same column descend from corresponding ancestral positions. This is why alignment is not merely formatting: it defines the character matrix that will be given to the phylogenetic model.

Insertions and deletions complicate this problem because modern sequences have different lengths. An alignment algorithm therefore balances residue similarity with gap penalties and, depending on the method, information from guide trees, pairwise alignments, or iterative refinement.

7.1 Progressive alignment

Progressive methods first estimate pairwise sequence relationships, build a guide tree, and progressively align sequences or profiles following that guide tree. They are fast and scalable, but early alignment decisions can propagate to later steps.

7.2 Iterative refinement

Iterative-refinement methods repeatedly split and realign parts of an existing alignment in an attempt to improve an objective score. This can reduce some of the path dependence of a purely progressive alignment.

7.3 MAFFT

MAFFT offers several strategies with different speed/accuracy trade-offs. Its name stands for Multiple Alignment using Fast Fourier Transform, reflecting one of the original innovations of the method. In protein sequences, amino acids can be represented numerically according to physicochemical properties, particularly volume and polarity. MAFFT can use the Fast Fourier Transform (FFT) to rapidly detect offsets at which two sequences show strong physicochemical correlation, thereby identifying candidate homologous regions without testing every possible positional shift individually. In other words, MAFFT can temporarily treat a protein sequence not only as a string of amino-acid letters, but also as a numerical signal whose properties can be compared using signal-processing methods.

MAFFT now includes several alignment strategies that differ substantially in how they construct and refine the final alignment. In this workshop, the choice between G-INS-i and L-INS-i will depend on the nature of the sequences being compared.

For extracted homologous domains that are expected to be globally alignable over most of their length, I generally prefer G-INS-i. G-INS-i uses global pairwise alignment information together with iterative refinement and is well suited to sequences of broadly similar length that share homology across their entire extent.

L-INS-i, in contrast, is particularly useful when sequences contain strongly conserved local regions separated by more variable segments, long insertions, or poorly alignable regions. It uses local pairwise alignment information together with iterative refinement and a consistency-based scoring scheme.

Both approaches are computationally more demanding than the faster MAFFT strategies, but this is generally not limiting for the relatively small datasets used in this workshop.

Typical workshop command:

```
# G-INS-i — preferred for extracted, globally homologous domains
mafft --globalpair --maxiterate 1000 input.fasta > alignment_ginsi.fasta
# L-INS-i — useful when local conserved regions are separated by variable regions
mafft --localpair --maxiterate 1000 input.fasta > alignment_linsi.fasta
```

7.4 MUSCLE 5

MUSCLE 5 emphasizes alignment ensembles. Rather than assuming that one deterministic alignment captures all plausible positional homology, ensemble approaches deliberately generate alternative high-quality alignments with different biases. The stability of downstream conclusions across those alternatives can be used as an additional measure of robustness.

7.5 What makes an alignment suspicious?

- Conserved catalytic or structural residues are displaced without a biological explanation.
- Long gap blocks occur inside clearly conserved structural cores.
- One highly divergent sequence forces many new gaps into all other sequences.
- Different algorithms produce incompatible alignments in the same region.
- Alignment columns are dominated by missing data or poorly supported positional equivalence.

Important

There is no universally correct trimming threshold. Removing ambiguous regions can reduce noise, but aggressive trimming can also remove genuine evolutionary information. The decision should be justified, not automatic.

8. Alignment quality and sensitivity analysis

Phylogenetic uncertainty begins before tree reconstruction. If an evolutionary conclusion changes drastically when MAFFT is replaced by another reasonable alignment, the uncertainty belongs in the biological interpretation.

MAFFT_ScoreNGo is used in this workshop to compare alignments using complementary metrics. The pedagogical goal is not to reduce alignment quality to one magic number, but to identify regions or sequences for which positional homology is unstable.

- Compare more than one reasonable alignment strategy.
- Inspect the alignment visually.
- Use quantitative metrics to flag unstable regions.
- Record any manual exclusion or trimming decision.
- If possible, test whether major clades are stable across plausible alignments.

9. Models of amino-acid evolution

Phylogenetic likelihood requires a model describing how amino acids change along branches. Protein substitution models encode the relative rates of different amino-acid replacements and the equilibrium frequencies expected over long evolutionary time.

Empirical matrices such as JTT, WAG and LG were estimated from large collections of protein alignments. More specialized matrices exist for particular biological contexts, and modern mixture models can represent additional heterogeneity.

9.1 The rate matrix Q

These evolutionary models are based on a continuous-time Markov process described by an instantaneous rate matrix, Q. For two different amino acids, i and j, the off-diagonal entry q_{ij} describes the instantaneous rate at which amino acid i is replaced by amino acid j. The diagonal entries are defined so that each row of the matrix sums to zero. Over a branch of length t, the rate matrix is converted into a matrix of substitution probabilities according to $P(t) = \exp(Qt)$. In phylogenetics, t usually represents the expected amount of evolutionary change along a branch, rather than chronological time itself. When Q is normalized so that the average substitution rate is one, branch lengths are expressed as expected substitutions per site. In commonly used time-reversible amino-acid models, these rates are determined by both the relative exchangeability between pairs of amino acids and their equilibrium frequencies.

9.2 Frequencies and rate heterogeneity

Real proteins are heterogeneous. Some sites are strongly constrained, whereas others evolve rapidly. Models can incorporate empirical amino-acid frequencies (+F) and among-site rate heterogeneity, for example with discrete gamma categories (+G) or FreeRate categories (+R).

Model notation therefore combines several assumptions. A label such as LG+F+R5 indicates an LG exchangeability matrix, empirical amino-acid frequencies and five FreeRate categories.

9.3 ModelFinder

ModelFinder evaluates candidate substitution models and compares their fit while penalizing model complexity. It calculates information criteria including AIC, AICc and BIC; the standard workflow commonly uses BIC for selection.

Model selection does not certify that the chosen model is biologically 'true'. It selects the best-supported option among the models that were considered.

10. Maximum-likelihood phylogenetic inference

Maximum likelihood asks: given an alignment and an evolutionary model, which tree topology, branch lengths and model parameters make the observed data most probable?

In compact notation: $L(T, \theta \mid A) = P(A \mid T, \theta)$

Here, A is the alignment, T is the tree topology, and θ represents branch lengths and model parameters. In practice, software maximizes the log-likelihood because products of many small probabilities are numerically inconvenient.

10.1 Likelihood is calculated site by site

For each alignment column, the ancestral amino acids are unknown. The likelihood calculation therefore sums over possible ancestral states at internal nodes. Efficient dynamic programming, classically the pruning algorithm, avoids enumerating every possible ancestral assignment explicitly.

10.2 Searching tree space

The number of possible unrooted bifurcating trees grows explosively with the number of taxa. For realistic datasets, exhaustive search is impossible. IQ-TREE therefore uses heuristic tree-search strategies that explore promising regions of tree space and optimize topology, branch lengths and model parameters.

Interpretation

The maximum-likelihood tree is the best tree found under the specified model and search procedure. It is not a proof that no other evolutionary scenario is possible.

10.3 Rooted versus unrooted trees

Standard reversible sequence models infer an unrooted topology. A root must be added using additional information or assumptions, for example a defensible outgroup, midpoint rooting, or another rooting method.

The root changes the direction of the evolutionary narrative. Therefore, rooting should be justified explicitly rather than treated as a cosmetic visualization step.

11. Branch support: SH-aLRT and ultrafast bootstrap

Branch support asks how strongly the data and analysis support a particular split. Different support statistics answer related but non-identical questions, which is why reporting more than one can be useful.

11.1 SH-aLRT

The SH-like approximate likelihood ratio test evaluates **whether a branch is supported relative to nearby alternative arrangements**. It is computationally efficient and provides a local measure of support.

11.2 Ultrafast bootstrap

Bootstrap approaches resample alignment columns to ask **how consistently a clade is recovered when the dataset is perturbed**. UFBoot accelerates this logic substantially relative to standard non-parametric bootstrapping.

For single-gene trees, IQ-TREE documentation commonly uses SH-aLRT ≥80 and UFBoot ≥95 as practical heuristics for strong support. These are not universal laws, and high support can still occur under systematic error such as model misspecification or alignment bias.

Do not overinterpret support

A branch can be very strongly supported and still be wrong if the data or model contain a systematic bias. Support measures quantify consistency under a procedure; they do not validate every assumption upstream.

12. Tree visualization is part of the analysis

iTOL and FigTree do more than make attractive figures. Mapping metadata onto the tree often reveals biological patterns, unexpected relationships, and possible annotation or sampling problems.

- Color branches or labels by taxonomy.
- Add domain architectures or protein lengths.
- Mark experimentally characterized proteins.
- Display branch-support values selectively.
- Distinguish rooted from unrooted representations.
- Avoid collapsing uncertainty into a visually definitive story.

13. Why use protein structure for deep evolutionary relationships?

Protein sequence can diverge rapidly, whereas the geometry of a functional fold is often constrained for longer evolutionary periods. At large evolutionary distances, structural comparison can therefore retain detectable relationships after sequence similarity becomes difficult to recognize.

This does not mean that structure is immune to convergence. Similar physical constraints can sometimes produce similar structural solutions independently, especially at coarse structural levels.

14. Foldseek and the 3Di structural alphabet

Foldseek makes large-scale structural comparison fast by converting local tertiary interactions into a sequence-like representation called the 3Di alphabet. For each residue, the structural state describes the geometry of its interaction with a spatially close residue.

Once structures have been encoded as 3Di strings, fast sequence-search machinery can be used as a prefilter. High-scoring candidates are then aligned using structural and amino-acid information.

Representation	What is compared?	Strength
Amino-acid sequence	Residue identities/substitutions	Very fast; rich evolutionary models
3Di structural alphabet	Local tertiary interaction states	Retains structural information in string form
3D superposition metrics	Coordinates / geometric similarity	Direct structural comparison but often more computationally expensive

Why Foldseek helps

Foldseek does not simply 'replace BLAST'. It answers a related but different question by searching structural similarity and is particularly useful when sequence-only detection of remote homologues becomes weak.

15. FoldTree: protein structural phylogenetics

FoldTree is a structural-phylogenetics pipeline that uses Foldseek for all-versus-all comparison of homologous protein structures. The published benchmark found that the strongest FoldTree approach used structural information to improve residue alignment and then derived a statistically corrected distance from sequence identity measured after the structure-informed alignment.

In the FoldTree pipeline, pairwise comparisons are compiled into a distance matrix and a distance-based tree is constructed. This is methodologically different from the maximum-likelihood sequence tree, so the two trees should be viewed as complementary analyses rather than two implementations of the same model.

FoldTree can also restrict comparisons to a conserved structural core, which is important when proteins contain lineage-specific additional domains.

15.1 Why can structural phylogenetics help?

- Structure often evolves more slowly than primary sequence.
- Structure-informed alignment can improve identification of equivalent residues.
- Very remote homologues may remain structurally recognizable.
- Structural evidence provides a complementary axis for testing sequence-based hypotheses.

15.2 Important limitations

- Structural convergence can mimic common ancestry.
- Predicted structures are models and contain variable confidence.
- Multidomain proteins can produce misleading pairwise distances if non-equivalent regions are compared.
- Distance-based structural trees and ML sequence trees have different statistical interpretations.
- Current structural phylogenetics does not eliminate the need for taxon sampling, curation or biological context.

Workshop rule

If the sequence and structural trees disagree, do not immediately choose one as correct. First identify which assumptions differ and what additional evidence could discriminate between explanations.

16. Comparing sequence and structural phylogenies

Observation	Possible interpretation	What to test next
Both trees recover the same major clade	Complementary signals are congruent	Check support, sampling and rooting
Sequence tree weak; structural tree stable	Sequence erosion or alignment difficulty may be limiting	Inspect MSA; add taxa; verify structural core
Sequence tree strong; structural tree different	Structural convergence, model differences or domain effects may matter	Inspect structures/domain boundaries; test alternative structural subsets
Both trees unstable	Dataset may not contain enough reliable signal	Revisit sampling, homology, domains and alignment
One taxon moves dramatically	Long branch, fragment, annotation error or paralogy	Inspect that sequence/structure individually

17. From a tree to an evolutionary hypothesis

A phylogeny becomes biologically useful when it is combined with independent evidence. Taxonomy, domain architecture, biochemical function, gene neighborhood, subcellular localization, structural features and experimental data can all constrain the plausible evolutionary narrative.

In discussion sections, distinguish clearly between what is directly shown by the analysis and what is an interpretation.

- Observation: 'Proteins A and B form a strongly supported clade.'
- Inference: 'This topology is consistent with a duplication before the divergence of lineages X and Y.'
- Alternative: 'Horizontal transfer followed by differential loss could also explain the pattern.'
- Test: 'Broader taxon sampling or species-tree reconciliation would discriminate between these scenarios.'

18. Reproducibility and reporting

A reproducible phylogenetic analysis should allow another researcher to reconstruct the dataset and rerun every major step.

Component	Minimum information to report
Sequences	Database, access date, accession list, taxon-selection criteria
Domains	Annotation resource, coordinates, extraction criteria
Alignment	Software version, algorithm/options, trimming/manual edits
Alignment QA	Metrics or sensitivity analyses used
ML phylogeny	IQ-TREE version, selected model, support methods and replicates
Rooting	Method and biological justification
Structures	Source, confidence filtering, region analyzed
Structural phylogeny	Foldseek/FoldTree versions and settings
Visualization	Software and metadata mapped onto tree
Availability	Repository, scripts, input files and final trees

19. Minimal command logic used in the workshop

The exact repository scripts will be provided separately. The following commands illustrate the main public-tool steps.

MAFFT G-INS-i (or alternatively MAFFT L-INS-i)

```
# G-INS-i — preferred for extracted homologous domains
mafft --globalpair --maxiterate 1000 domain.fasta > alignment_ginsi.fasta
# L-INS-i — useful when conserved regions are separated by variable segments or long insertions
mafft --localpair --maxiterate 1000 domain.fasta > alignment_linsi.fasta
```

IQ-TREE 3: model selection + ML tree + SH-aLRT + UFBoot

```
iqtree3 -s alignment.fasta -m MFP -alrt 1000 -B 1000 -T AUTO
```

Foldseek: optional structural search example

```
foldseek easy-search query.pdb target_db hits.tsv tmp/
```

FoldTree will be run through the prepared workshop workflow rather than reconstructed manually from individual Foldseek commands.

20. Short glossary

Term	Meaning
Alignment column	A set of residues hypothesized to descend from corresponding ancestral positions.
Branch length	Model-dependent estimate of evolutionary change along a branch.
Homologue	A gene, protein, domain, or other biological feature related to another by common ancestry.
Orthologue	Homologues related by a speciation event.
Paralogue	Homologues related by a duplication event.
Topology	The branching relationships in a tree, independent of branch lengths.
Substitution model	A probabilistic model of how molecular characters change through time.
Maximum likelihood	An inference principle that chooses parameter values maximizing the probability of the observed data under a model.
UFBoot	Ultrafast bootstrap approximation used to assess clade stability under site resampling.
SH-aLRT	Approximate likelihood-ratio support test for individual branches.
3Di	Foldseek structural alphabet encoding local tertiary residue interactions.
pLDDT	AlphaFold per-residue confidence metric.
PAE	Predicted Aligned Error; confidence metric for relative positioning of residues/regions.

21. Recommended reading and documentation

1. UniProtKB documentation. https://www.uniprot.org/help/uniprotkb
2. InterPro / InterProScan documentation. https://www.ebi.ac.uk/interpro/
3. AlphaFold Protein Structure Database FAQ. https://alphafold.ebi.ac.uk/faq
4. Katoh K, Standley DM. MAFFT Multiple Sequence Alignment Software Version 7: Improvements in Performance and Usability. Mol Biol Evol. 2013;30:772–780. doi:10.1093/molbev/mst010.
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14. Puente-Lelievre C, Malik A, Douglas J. Protein Structural Phylogenetics. Genome Biol Evol. 2025;17:evaf139. doi:10.1093/gbe/evaf139.
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22. Take-home message

Deep phylogeny is not one algorithm. It is a chain of biological and statistical decisions: define the question, sample taxa intelligently, compare homologous regions, evaluate the alignment, choose an evolutionary model, quantify support, inspect structural evidence, and challenge the resulting narrative with alternative explanations.

The most useful result of the workshop is therefore not a single tree. It is the ability to recognize which parts of an evolutionary inference are robust, which depend on assumptions, and what experiment or analysis should come next.